

Education

Ph.D. in Computer Science , University of Waterloo, Canada	Sept. 2021 — Nov. 2025
- Advisors: Dr. Naomi Nishimura and Dr. Amer E. Mouawad - Thesis: “On Token Movement Problems”	
B.Sc. in Computer Science , American University of Beirut, Lebanon	Jan. 2019 — May 2021
Graduated with High Distinction, GPA: 4.0/4.0	
B.B.A. - with a focus on Finance, American University of Beirut, Lebanon	Sept. 2013 — Dec. 2016

Scholarships and Awards

Computer Science PhD Student Award , University of Waterloo [3,000 CAD]	2025 — 2026
International Doctoral Student Award , University of Waterloo [60,000 CAD]	2022 — 2025
Mathematics International Master’s Award of Excellence , University of Waterloo [7,500 CAD]	2021 — 2022
University of Waterloo Graduate Scholarship , University of Waterloo [4,000 CAD]	2021 — 2022
Henri Qais Naccache Computer Science Student Development Fund , American University of Beirut [500 USD]	2020 — 2021
United States AID University Scholar , American University of Beirut [125k+ USD]	2013 — 2016

Conference Publications (*Alphabetical Authorship)

[ISAAC’24]*	Mario Grobler, Stephanie Maaz , Amer E. Mouawad, Naomi Nishimura, Vijayaragunathan Ramamoorthi, and Sebastian Siebertz. Kernelization Complexity of Solution Discovery Problems , The 35th International Symposium on Algorithms and Computation, 2024 [pdf]
[ICALP’24]*	Mario Grobler, Stephanie Maaz , Nicole Megow, Amer E. Mouawad, Vijayaragunathan Ramamoorthi, Daniel Schmand, and Sebastian Siebertz. Solution Discovery via Reconfiguration for Problems in P , The 51st EATCS International Colloquium on Automata, Languages, and Programming, 2024 [pdf]
[WALCOM’22]*	Alexandre Cooper, Stephanie Maaz , Amer E. Mouawad, and Naomi Nishimura. Parameterized Complexity of Reconfiguration of Atoms , The 16th International Conference and Workshops on Algorithms and Computation, 2022 [pdf]

Journal Publications (*Alphabetical Authorship)

[Algorithmica’24]*	Alexandre Cooper, Stephanie Maaz , Amer E. Mouawad, and Naomi Nishimura. Parameterized Complexity of Reconfiguration of Atoms , Algorithmica, 2024 [pdf]
[PRA’23]	Remy El Sabeh, Jessica Bohm, Zhiqian Ding, Stephanie Maaz , Naomi Nishimura, Izzat El Hajj, Amer E. Mouawad, and Alexandre Cooper. Efficient Algorithms to Solve Atom Reconfiguration Problems. II. Assignment-Rerouting-Ordering Algorithm , Physical Review A, 2023 [pdf]
[PRA’23]	Barry Cimring, Remy El Sabeh, Marc Bacvanski, Stephanie Maaz , Izzat El Hajj, Naomi Nishimura, Amer E. Mouawad, and Alexandre Cooper. Efficient Algorithms to Solve Atom Reconfiguration Problems. I. Redistribution-Reconfiguration Algorithm , Physical Review A, 2023 [pdf]

Preprints (*Alphabetical Authorship)

2025*	Nicolas Bousquet, Stephanie Maaz , Amer E. Mouawad, Naomi Nishimura, and Sebastian Siebertz. On Algorithmic Meta-Theorems for Solution Discovery: Tractability and Barriers , Preprint, arXiv:2510.17344, 2025 [pdf]
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Academic Service

External Reviewer	Workshop on Graph-Theoretic Concepts in Computer Science [WG'23] International Colloquium on Automata, Languages, and Programming [ICALP'25]
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Contributed Talks

Nov. 2024	Kernelization Complexity of Solution Discovery Problems, Mohamed bin Zayed University of AI (MBZUAI) , Abu Dhabi, UAE
Jan. 2024	Parametrized Complexity of Reconfiguration of Atoms, 1st Villeurbanne Reconfiguration Workshop (VRW'24) , Lyon, France
May 2022	Parametrized Complexity of Reconfiguration of Atoms, Banff International Research Station (BIRS'22) , Banff, Canada

Service

2023 — 2025	Senator, Graduate Student Representative , University of Waterloo, Canada Participated in university academic governance (including graduate research regulations); liaised with graduate student bodies
2015 — 2017	Community Service Organizer , U.S. Agency for International Development , Lebanon Worked with a team to secure funding to open an ancient cultural wilderness trail; collaborated with local communities (mapping the trail, clearing pathways, installing signage); documented the project in reports
2013 — 2016	Community Service Volunteer , CCECS, American University of Beirut , Lebanon Tutored over 15 refugees at the NASMA Learning and Resource Center to improve academic and social skills
2014	Community Service Volunteer , D-RASATI - Education Development Center , Lebanon Participated in a nationwide assessment (300+ Lebanese public schools) aimed at equipping computer and science laboratories; interviewed principals; evaluated technological readiness

Experience

Teaching Assistant , University of Waterloo, Canada <i>Courses</i> : CS245 Logic and Computation, CS251 Computer Organization & Design	Sept. 2021 — Aug. 2024
Financial Analyst , PFC International , Lebanon <i>Focus</i> : Strategic planning & advisory, investment analysis, and market research	Oct. 2017 — Dec. 2018
Intern , Business Engineering Experts , Lebanon <i>Focus</i> : Corporate governance, financial model design, and cost of equity estimation	May. 2017 — Jun. 2017
Intern , Byblos Bank Group , Lebanon <i>Focus</i> : Compliance with key regulatory objectives such as FATCA and confidentiality	Jun. 2016 — Jul. 2016

References

Naomi Nishimura	University of Waterloo	nishi@uwaterloo.ca
Amer E. Mouawad	American University of Beirut	aa368@aub.edu.lb
Sebastian Siebertz	University of Bremen	siebertz@uni-bremen.de
Nicolas Bousquet	CNRS, LIRIS, University of Lyon	nicolas.bousquet@cnrs.fr